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RESEARCH-ARTICLE

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A Novel Method for Extracting Frequency Line on Lofargram based on Feature Function

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ABSTRACT

Passive sonar signals are generated by receiving and processing the complex noise of the surrounding environment, so it is very difficult to extract useful information from them. In this paper, a novel algorithm for extracting frequency lines on lofargram based on feature function is proposed. Firstly, the lofargram is preprocessed before extracting the frequency line, which makes the frequency line in the lofargram more obvious. Secondly, according to the high brightness, continuity and smoothness of frequency line information, density based spatial clustering of applications with noise (DBSCAN) and polynomial fitting are used to set the detection window for feature function. Finally, simulation experiments and real data experiments verify that the proposed algorithm also has good performance in low signal-to-noise ratio lofargram.

CCS CONCEPTS

• Computing methodologies; • Computer graphics; • Image manipulation; • Image processing; • Machine learning; • Learning paradigms; • Unsupervised learning;

KEYWORDS

passive sonar, lofargram, image processing, DBSCAN, frequency line detection

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1 INTRODUCTION

Ships, torpedoes and other targets will produce radiated noise when sailing. Passive sonar signals are generated by receiving and processing radiated noise or sonar signals from underwater targets [1]. Lofargram can be obtained by short-time Fourier transform (STFT) of the passive sonar signal, and its signal-to-noise ratio (SNR) is often low. In practical applications, spectrum analysis is usually carried out by lofargram, which is called low frequency analysis recording (LOFAR). Frequency line in lofargram is very important

for target detection, tracking and motion state estimation in the ocean [2]. The research on extracting frequency line of lofargram mainly focuses on optimization methods [3-5], image processing methods [6, 7], hidden Markov method [8, 9], Statistical methods [10, 11] and so on. The feature function method is an optimization method. Di Martino proposed the feature function of the frequency line in lofargram, and proposed a frequency line detection method based on the feature function [4]. The algorithm can reduce noise and improve the detection probability of the frequency line. An improved feature function is proposed, and then a frequency line feature cumulative detection method with a sliding window in the frequency domain is proposed [5]. In this method, a moving narrow-band observation window is set in the lofargram, and the optimal path is found according to the frequency line feature function in the observation window. Simulation results show that this method has good detection ability for frequency line signals with time-varying frequency and low SNR. Moreover, it can enhance and detect multiple frequency lines.

The feature function method considers the gray value of lofargram, the continuity of frequency line and the frequency stability. In recent years, image processing and machine learning methods have attracted more and more attention. In view of the low SNR of lofargram and the gray value of frequency line is slightly larger than the gray value of surrounding noise, before using the feature function to detect frequency line, this paper first uses the image processing method to preprocess lofargram, which makes the frequency line information more obvious. Then, using the results of the DBSCAN (density based spatial clustering of applications with noise) [12], curve fitting and image processing, a new frequency line extraction method based on feature function is proposed. The application of image processing and machine learning methods greatly improves the stability and universality of feature function in frequency line detection. The main flowchart is shown below.

The structure of this paper is as follows. In the second part, the lofargram preprocessing method for highlighting frequency lines is introduced. The third part introduces the feature function and the frequency line extraction algorithm based on the DBSCAN and curve fitting. The fourth part shows the performance of the proposed algorithm in simulated images and actual data. Finally, this paper is summarized and prospected.

2 IMAGE PREPROCESSING

Image preprocessing is the basis of image feature extraction and recognition [13]. Its purpose is to minimize all kinds of noise caused by environment, hardware and system, and highlight the points of frequency line in the image. The lofargram preprocessing steps are as follows.

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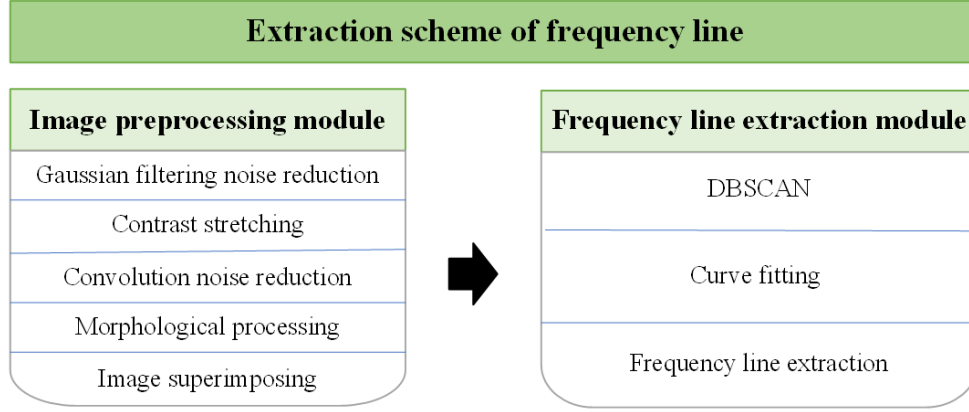


Figure 1: Flowchart of the proposed algorithm

2.1 Noise Reduction

2.1.1 Gaussian Filtering Noise Reduction. There are too many isolated scatter noises in lofargram. Firstly, Gaussian filtering is used to reduce the noise of the image, which can achieve a certain degree of smoothing effect.

2.1.2 Contrast Stretching. Through observation, it is easy to know that the frequency line in the image is often presented in the form of highlighting. If the gray value of a pixel in the image is small, it may be considered that the pixel is a noise point, and ignoring the low gray value has almost no effect on the trend and direction of the frequency line. In order to better highlight the frequency line points and facilitate subsequent detection, this paper adopts the linear contrast stretching method and sets the threshold of the contrast stretching method according to the average value of non-zero gray values in the image.

The mean value of non-zero gray value is defined as follows:

$$g_{mean} = \frac{\sum_{f=1}^{N_f} \sum_{t=1}^{N_t} g(f, t)}{\sum_{f=1}^{N_f} \sum_{t=1}^{N_t} I(g(f, t) \neq 0)} \quad (1)$$

Among them,

$$I(g(f, t) \neq 0) = \begin{cases} 1, & g(f, t) \neq 0 \\ 0, & g(f, t) = 0 \end{cases} \quad (2)$$

N_f, N_t Indicate the length of frequency and time respectively, that is, the length and width of the image, and $g(f, t)$ indicates the gray value of the image at the point.

The linear contrast stretching method used is as follows:

$$g(f, t) = \begin{cases} 0, & g(f, t) \leq a * g_{mean} \\ \frac{g(f, t) - a * g_{mean}}{(b - a) * g_{mean}}, & a * g_{mean} < g(f, t) \leq b * g_{mean} \\ 1, & g(f, t) \geq b * g_{mean} \end{cases} \quad (3)$$

a and b are constants here.

The contrast stretching method can highlight the line's points in the image and suppress the noise points in the image to a great extent. Moreover, the stretching threshold is determined according to the relationship between the gray value and the image mean

value, which can be applied to the contrast stretching of all lofargrams. After many experiments, it can be found that when a and b are changed in a small range, it has little influence on the image. So $a = 1.1, b = 2$ can be taken in the frequency line detection of passive sonar lofargram.

2.1.3 Convolution Noise Reduction. In order to improve the stability of the algorithm and adapt to more lofargrams, a convolution denoising operation is necessary. This paper uses the cross convolution kernel to convolve the image, and then retains the larger value of each row in the image, which can effectively remove the scattered interference in the lofargram.

2.2 Frequency Line Enhancement

2.2.1 Morphological Processing. To enhance the frequency line, the closing operation is first used. The closing operation is to expand first and then carry out etching operation, in which the expansion operation can make the tiny objects in the image bigger, while the etching operation can delete the small objects in the image. The closing operation can fill the interrupted frequency line in the image to make it continuous in the time dimension.

Then binary the data after the closing operation, and the binary function adopted in this paper is as follows:

$$g(f, t) = \begin{cases} 0, & g(f, t) \leq c \\ 1, & g(f, t) > c \end{cases} \quad (4)$$

Where c is a constant, which is called binarization threshold.

The frequency line is thin, so it is necessary to refine the above results. Skeleton extraction in morphology is a common thinning operation. But skeleton extraction will lead to the loss of frequency line information and produce many branches, which does not conform to the image characteristics of the frequency line. Therefore, this paper first finds the largest connected domain in the above figure and sets its area as S_{max} . If the area of a connected domain is greater than $0.05 \times S_{max}$, the connected domain is retained, and the mean value of coordinates of each row of the connected domain is calculated. Then the thinned image of the binarization result can be obtained, which is called skeleton point, but it is different from the skeleton point in traditional image processing.

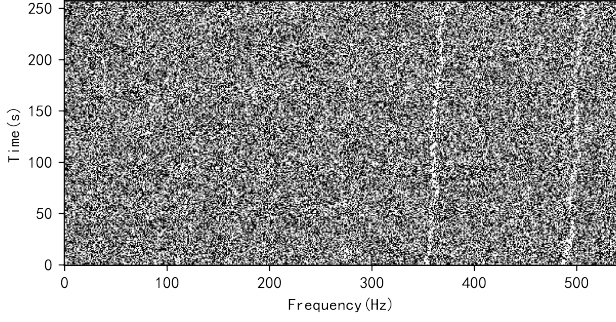


Figure 2: Original image of the simulated lofargram

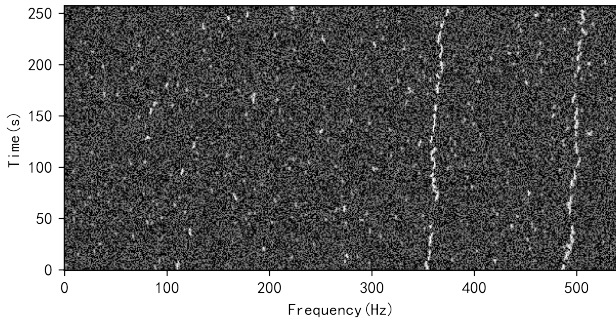


Figure 3: lofargram with image preprocessing

2.2.2 Superimposing. By superimposing the above processing results with the original image according to a certain proportion, and then normalizing the superimposed image, the frequency line in the original image can be more obvious. At the same time, the frequency line points deleted in the previous preprocessing steps can be restored to a certain extent.

At this point, the image preprocessing has been completed. Figure 2 shows a simulated lofargram. Figure 3 is a lofargram after the image preprocessing steps described above. It can be seen that image preprocessing can enhance the bright areas in the image while weakening the darker areas in the image. That is, image preprocessing can enhance the target frequency line and reduce the interference of environmental noise. The frequency line data in the pretreated image is more obvious, which plays a vital role in improving the accuracy of subsequent frequency line detection.

3 FREQUENCY LINE EXTRACTING

3.1 Feature Function

The frequency line component in lofargram has the basic characteristics of local maximum, continuous and slow change of frequency line intensity. However, the intensity of background noise in lofargram is random, and the bright spots formed by it are not continuous in time and frequency. [5] designed the feature function according to the above characteristics of lofargram, as follows:

$$\varphi(\xi) = \frac{A(\xi)}{\alpha G(\xi) + \beta C(\xi) + \gamma} \quad (\gamma \neq 0) \quad (5)$$

Among them, ξ represents an integral path in the observation window of lofargram. The path length (also the length of the time) is set to N_t . Each pixel point on the path is set to P_t ($1 \leq t \leq N_t$). $A(\xi), C(\xi), G(\xi)$ stand for intensity characteristics, frequency continuity and trajectory continuity characteristics of frequency line respectively. α, β, γ are three weight coefficients. The definitions of $A(\xi), C(\xi), G(\xi)$ are as follows:

$$A(\xi) = \sum_{t=1}^{N_t} g(P_t) \quad (6)$$

$$d(P_t, P_{t+1}) = f(P_t) - f(P_{t+1}) \quad (7)$$

$$C(\xi) = \sum_{t=1}^{N_t} |d(P_{t-1}, P_t) - d(P_t, P_{t+1})| \quad (8)$$

$$h(P_t) = \begin{cases} 1, & g(P_t) = 0 \\ 0, & g(P_t) \neq 0 \end{cases} \quad (9)$$

$$G(\xi) = \sum_{t=1}^{N_t} h(P_t) \quad (10)$$

The amplitude of P_t is $g(P_t)$, and its frequency value is $f(P_t)$. $d(P_t, P_{t+1})$ indicates the frequency gradient of two points at adjacent times of the path. $h(P_t)$ is a marked variable. If P_t is a breakpoint, it is recorded as 1, otherwise it is 0. $g(P_t) = 0$ indicates that there is a breakpoint here. According to the actual situation, the point whose amplitude is lower than a certain threshold in the lofargram can be set as a breakpoint, and the threshold value is generally set as the average value of the lofargram.

When the path approaches or passes through the frequency line, the sum of amplitudes of points on the path increases, the number of segments decreases, the frequency difference decreases, that is, $A(\xi)$ increases, $C(\xi)$ and $G(\xi)$ decreases, and the feature function $\varphi(\xi)$ increases. In the observation window, the path when it is the maximum value is the optimal path.

Using the feature function, [5] also proposed the cumulative detection method of frequency line features with a sliding window in frequency domain. In this method, a moving narrow-band observation window is set in the lofargram. In the window, the path with the maximum feature function is taken as the frequency line to be detected. Move the narrow-band observation window to get the optimal path of each position observation window. Finally, the number of lofargram pixels passing by the optimal path is accumulated to obtain the frequency line enhancement map.

3.2 Proposed Algorithm

The method of extracting frequency lines based on feature functions can well identify some frequency lines in lofargram. The result of the feature function detection depends not only on the image quality, but also on the position and size of the window. Also, if the position and size of the window are fixed, it is difficult to detect the frequency change of the frequency line when using the feature function to extract the frequency line. Therefore, we use the DBSCAN, polynomial fitting, and skeleton points to propose a new method of using feature function to extract frequency lines.

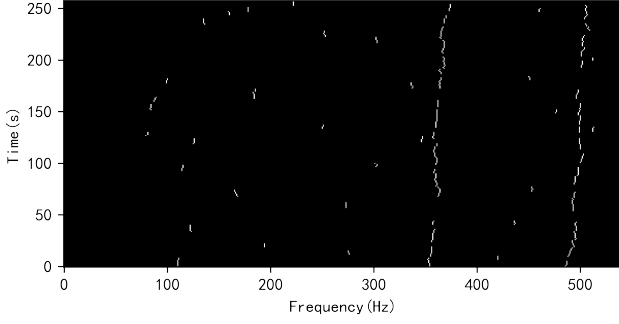


Figure 4: The noisy skeleton point of the frequency line

3.2.1 Calculation of the Center Frequency. In the image preprocessing results, the skeleton points are closest to the frequency line information (as shown in Figure 4). Therefore, the coordinate information of skeleton points is clustered by the DBSCAN. DBSCAN is a density clustering method, which does not need to specify the number of target categories, and can judge the number of categories according to the density of data [12].

When the skeleton points are clustered by the DBSCAN, the points of the same class represent the points within the same frequency line. Because there are still some noise points in the skeleton points obtained in the third section, it is necessary to filter the clustering results. If the sample number of a certain class is less than the specified threshold, it is considered that the class is the result of clustering noise points, and it needs to be eliminated. The number of classes whose sample number is greater than the specified threshold represents the number of frequency lines in the graph.

The average frequency of all points of a certain class whose sample number is larger than the threshold is considered as the frequency center of its corresponding frequency line, and also as the starting frequency of frequency line extraction. It is reliable to regard the frequency mean as the center of the frequency line, because the noise points are random, so the distribution of outliers around the frequency line is random.

By fitting all points in the class with quartic polynomial, a smooth polynomial curve almost consistent with the frequency line can be obtained. Taking the frequency value of the fitting curve at one time as the window center frequency when detecting the frequency line at the next time, can help the feature function detect the frequency change of the frequency line.

Taking the skeleton points in figure 4 as an example, the results of polynomial fitting of points in the categories that meet the threshold requirements are shown in figure 5

From Figure 5, we know that after the DBSCAN, two categories meet the threshold requirements, and their corresponding frequency averages are 361 Hz and 498 Hz, which means that there are two lines in the graph, and their frequency line extracting starting frequencies are 361 Hz and 498 Hz respectively. Combined with the information in the graph, it can be seen that the starting frequencies determined by this method are reliable.

The starting frequency of frequency line detecting and the number of frequency lines are well determined by this method, which avoids the traversal operation of the algorithm in [5]. Moreover,

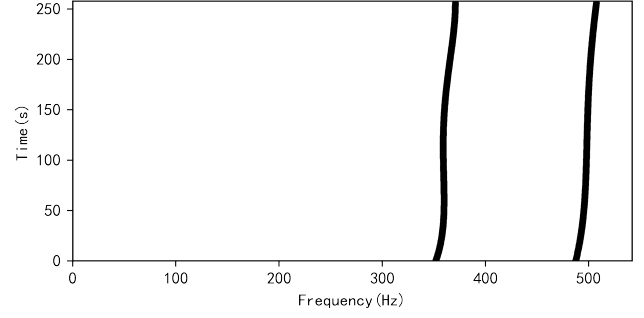


Figure 5: The fitted curve of skeleton point

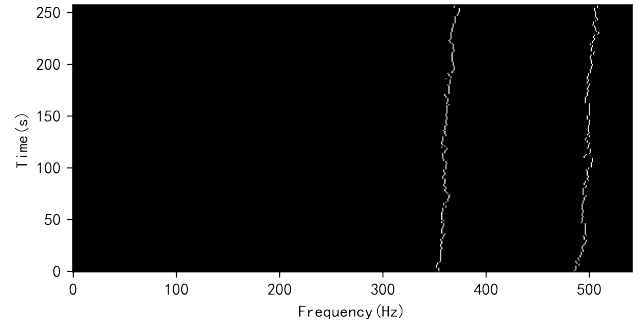


Figure 6: The frequency line extraction result obtained by the proposed algorithm

the window position of feature function detecting can be adjusted according to the fitting curve, which greatly improves the accuracy and stability of frequency line extraction.

3.2.2 Frequency Line Extraction. It can be found that the frequency change of skeleton points in image preprocessing is consistent with the frequency line data in the original image, so it is required to detect the frequency line through skeleton points as much as possible. Firstly, the skeleton point corresponding to the starting frequency is found as the starting point of frequency line extraction, and the extraction is started up and down along the time axis respectively.

Taking upward detecting as an example, let's set the frequency of fitting polynomial at time t to f_t . First, define two windows,

$$Win_1 = [f_t - win_1, f_t + win_1] \quad (11)$$

$$Win_2 = [f_t - win_2, f_t + win_2] \quad (12)$$

If skeleton points are included in Win_2 , and the frequency of skeleton points is f_t' , Win_1 will be corrected to $Win_1 = [f_t' - win_1, f_t' + win_1]$, otherwise it will remain unchanged. $win_2 > win_1$ is often chosen.

In order to find the frequency of the frequency line point at time $t + 1$, the value of the feature function at that time is first calculated, f is limited in Win_1 ,

$$\Phi((f, t + 1)) = \frac{g(f, t+1) - g(f, t)}{\alpha * |f - f_t| + \gamma}, \quad f \in Win_1 \quad (13)$$

Find the frequency f that maximizes $\Phi((f, t + 1))$ as the frequency at time $t + 1$, and set to f_{t+1} . That is,

$$f_{t+1} = \operatorname{argmax} \Phi((f, t + 1)), \quad f \in \text{Win}_1 \quad (14)$$

Based on this idea, the final frequency line can be obtained as follows:

The specific algorithm is as follows:

Algorithm 1 Frequency line extraction algorithm

labels: DBSCAN results in different categories of unique tags, and the number of points in this category is greater than a threshold (Default is 70).

lines[label]: After the DBSCAN, all skeleton points belonging to the label.

n_lines: The number of lines detected, and the initial value is 0

win₁: Window width. Default is 2

win₂: Window width. Default is 10

for *label* in *labels*, do

n_lines = *n_lines* + 1

$$f_0 = \frac{\sum_{(f,t) \in \text{lines}[\text{label}]} f}{n}$$

t_0 = The time of skeleton point corresponding to the f_0 ,

$f_{t_0} = f_0$

Polynomial fitting to $(f, t) \in \text{lines}[\text{label}]$, denoted $p(t)$

$t = t_0$

while $t \leq N_t - 1$, do

$\text{Win}_2 = [f_t - \text{win}_2, f_t + \text{win}_2]$

If there is skeleton point in Win_2 , do

Set the frequency of the skeleton point to f_t'

$$\text{Win}_1 = [f_t' - \text{win}_1, f_t' + \text{win}_1]$$

else:

$$\text{Win}_1 = [p(t) - \text{win}_1, p(t) + \text{win}_1]$$

$$\Phi((f, t + 1)) = \frac{g(f, t+1) - g(f, t)}{\alpha * |f - f_t| + \gamma}, \quad f \in \text{Win}_1$$

$$f_{t+1} = \operatorname{argmax} \Phi((f, t + 1)), \quad f \in \text{Win}_1$$

$t = t + 1$

end

$t = t_0$

while $t \geq 2$, do

$\text{Win}_2 = [f_t - \text{win}_2, f_t + \text{win}_2]$

If there is skeleton point in Win_2 , do

Set the frequency of the skeleton point to f_t'

$$\text{Win}_1 = [f_t' - \text{win}_1, f_t' + \text{win}_1]$$

else:

$$\text{Win}_1 = [p(t) - \text{win}_1, p(t) + \text{win}_1]$$

$$\Phi((f, t - 1)) = \frac{g(f, t-1) - g(f, t)}{\alpha * |f - f_t| + \gamma}, \quad f \in \text{Win}_1$$

$$f_{t-1} = \operatorname{argmax} \Phi((f, t - 1)), \quad f \in \text{Win}_1$$

$t = t - 1$

end

end

4 EXPERIMENTS AND RESULTS

For low SNR images, the literature [5] constructed a feature function and designed a frequency-domain sliding window frequency line feature accumulation detection method to detect the frequency line in the lofargram. In recent years, the application of image

processing and machine learning algorithms has greatly improved many traditional algorithms. In this paper, image processing and machine learning methods are added on the basis of the method in [5], which avoids the use of sliding windows and more accurately extracts frequency lines in lofargrams. The following shows the results of the algorithm in [5] and the algorithm in this paper on real data and simulated data with different noises added.

4.1 Simulation Data Experiment

According to the characteristics of lofargram noise, such as clutter and irregularity, we use random numbers, salt and pepper noise and Gaussian noise to simulate the frequency line image. In the figure below, (a) and (d) are simulated lofargrams with different levels of noise added, where (d) has a larger noise level than (a). (b) and (c) are the frequency line results in (a) detected by the algorithm in [5] and the algorithm proposed in this paper, respectively. (e) and (f) are the frequency lines in (d) detected by the algorithm in [5] and the algorithm proposed in this paper, respectively.

In the above results, the algorithm in this paper and the algorithm in the literature [5] perform well in Figure 7(a). However, on lofargram with low SNR (Figure 7(d)), the smoothness and completeness of the results in [5] are poor, while our results can still detect the general trend of the frequency line. This shows that the improvement of literature [5] in this paper is effective. Although the noise levels of (a) and (d) differ, the proposed algorithm can extract complete frequency lines, and the results match those observed with the naked eye. This demonstrates that the algorithm in this paper is stable and has significant importance for eye-to-eye observation of release workers.

4.2 Real Data Experiment

To verify the practicability of the algorithm in this paper, this paper also applies the proposed algorithm to the real data of sea trials. The data used is from a public dataset [14], which provides underwater sounds produced by various types of ships. These sounds were recorded in shallow water and real conditions. Therefore, the recording contains both natural environment noise and artificial environment noise. Some lofargrams were obtained by doing STFT for the voice in [14], and two of them were selected (Figure 8(a) and Figure 8(d)). The algorithm in the literature [5] and the algorithm proposed in this paper are used to detect the frequency lines in the two lofargrams, and the results are shown in Figure 8

The above results show that the algorithm proposed in this paper also performs well in the sea trial data. And by comparing with the results of literature [5], it can be found that the smoothness of the frequency line detected by the algorithm in this paper is better. This is due to the method of extracting skeleton points using the mean value designed in this paper in the image preprocessing step, which is smooth. At the same time, the skeleton point and the fitting curve are combined to continuously adjust the detection window of the characteristic function, which avoids the use of a sliding window and also makes the detected frequency line smoother.

5 SUMMARY AND PROSPECT

The proposed algorithm cleverly applies methods such as image processing, DBSCAN clustering and curve fitting. By continuously

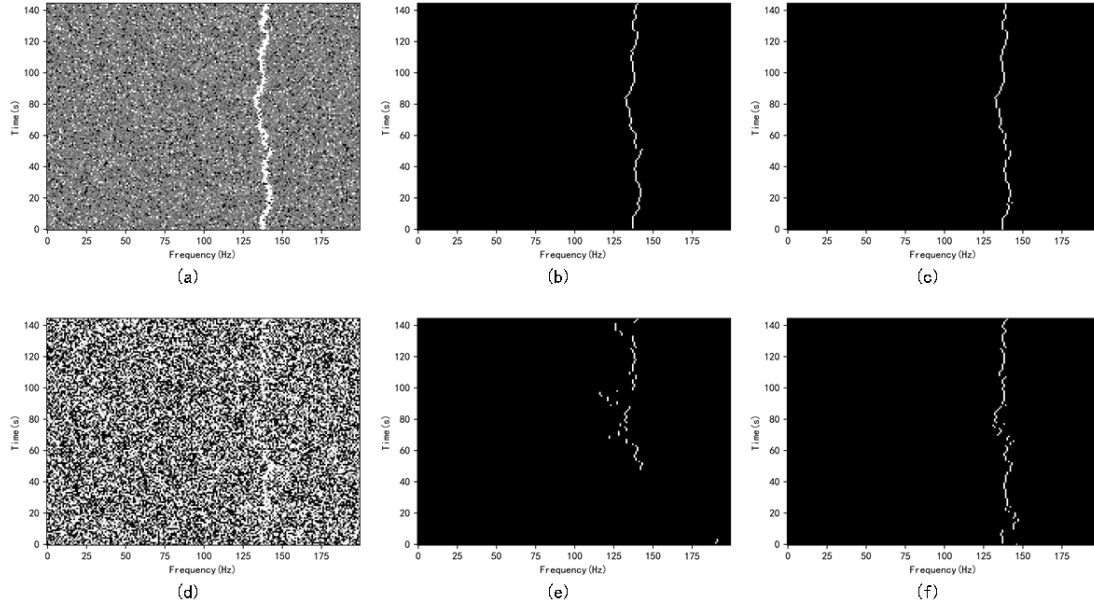


Figure 7: Simulation lofargrams of different noises and frequency line detection results: (a) The lofargram with less noise; (b) The line in (a) extracted by the algorithm in [5]; (c) The line in (a) extracted by the proposed algorithm; (d) The lofargram with more noise; (e) The line in (d) extracted by the algorithm in [5]; (f) The line in (d) extracted by the proposed algorithm.

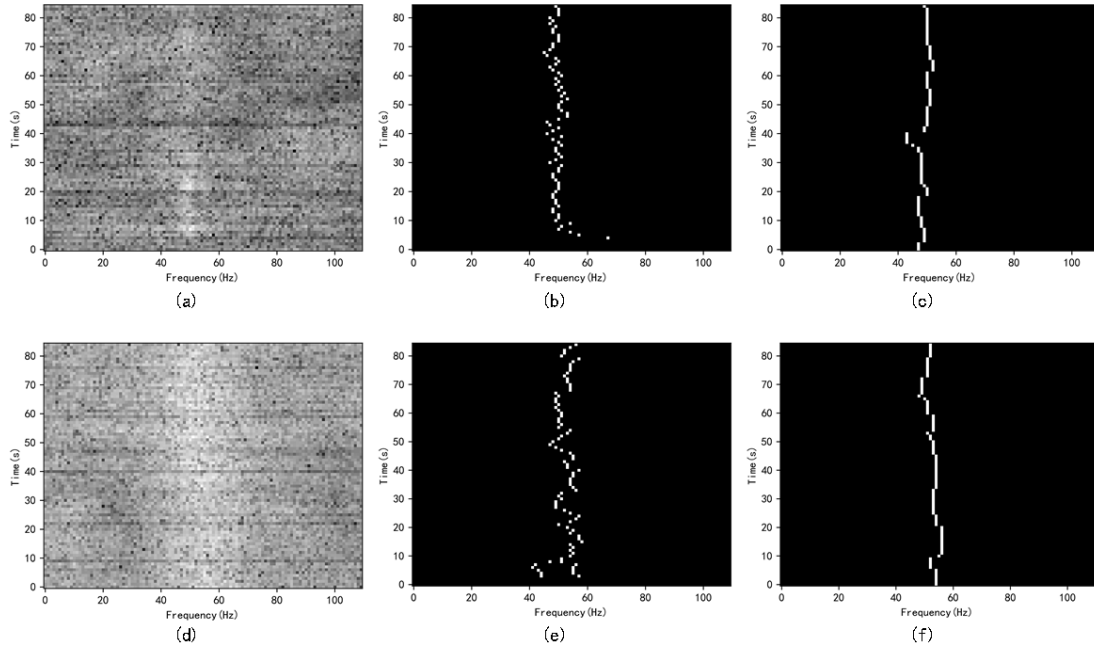


Figure 8: real lofargrams and frequency line detection results: (a) Lofargram 1; (b) The line in (a) extracted by the algorithm in [5]; (c) The line in (a) extracted by the proposed algorithm; (d) Lofargram 2; (e) The line in (d) extracted by the algorithm in [5]; (f) The line in (d) extracted by the proposed algorithm.

adjusting the window position and size of the feature function, the generality of the frequency line detection algorithm and the smoothness of the results are greatly improved. The algorithm helps computers identify the frequency line in the lofargram. Experiments show that the proposed algorithm can extract the frequency line from the lofargram and detect its fluctuations effectively. Furthermore, it performs well in real data and simulated data with different noises. It is of great significance to automatically detect the frequency lines in the lofargram, and then automatically identify whether there is a target around the sonar receiver and its motion status. In the future, we will use the frequency line extraction method proposed in this paper and neural network to estimate and classify objects in the ocean.

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